This booklet is part of a continuing patient education series on the thyroid gland and thyroid disease. Additional booklets available in this series are:

Getting to Know Your Medication
How the Thyroid Gland Can Affect Your Weight
Hypothyroidism and Your Health
Hyperthyroidism and Your Health
The Postpartum Period and Your Thyroid
The Thyroid Gland and Your Health
Thyroid Disease and Aging
Thyroid Disease and Diabetes
Thyroid Disease in Women
Understanding Goiter
Understanding Graves' Disease
Understanding Thyroid Nodules
Understanding TSH Testing
Understanding Your Child's Thyroid
What Is Hashimoto's Disease?

Compliments of Knoll Pharmaceutical Company

knoll®

Knoll Pharmaceutical Company
3000 Continental Drive - North
Mount Olive, NJ 07828-1234

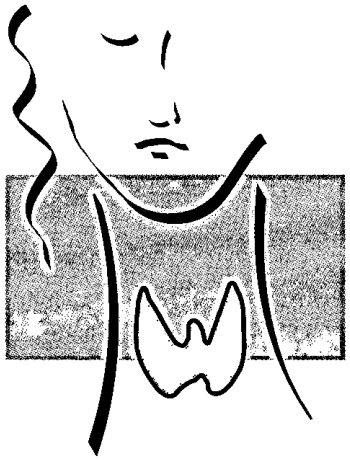
BASF Pharma

© 1998 Knoll Pharmaceutical Company
150M

All rights reserved.
SYN-0298-075/5-98

Printed in USA
EDP 9990209

Understanding Thyroid Cancer



Thyroid Cancer— What You Need to Know

Thyroid cancer is a malignant (harmful) lump or growth located within the thyroid gland. Fortunately, thyroid cancer is very *rare*, and in most patients grows very slowly. Furthermore, the outlook with treatment is excellent—most thyroid cancers can be totally removed with surgery and a complete cure is common.

The following pages contain basic information about the thyroid gland, its role in your body, and thyroid disorders. In addition, this booklet will discuss the diagnosis and treatment of thyroid cancer.

Remember, no single booklet can answer all of your questions or replace speaking with doctors, nurses, and other members of the healthcare team. If you have any questions or concerns about the thyroid or thyroid disorders, please check with your doctor.

Notes:

Notes:

Contents

Your Thyroid—The Basics....4

Thyroid Disorders—A Brief Look at What Can Go Wrong....7

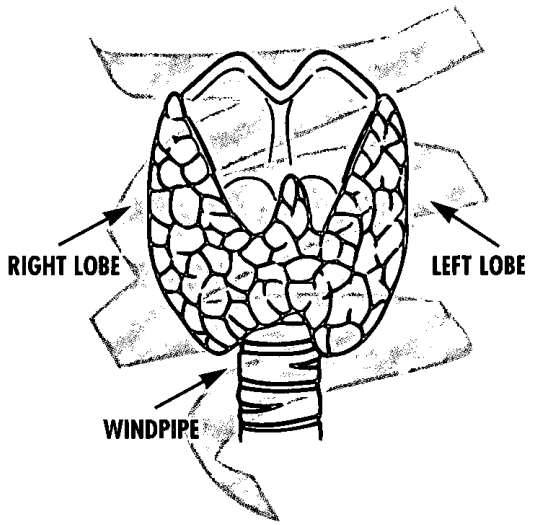
Thyroid Cancer—A Closer Look....10

- Who Is at Risk?....11
- Diagnosis of Thyroid Cancer....12
- Treatment of Noncancerous Thyroid Nodules....17
- Suspicious and Nondiagnostic Nodules....18
- Cancerous Thyroid Nodules....19

Doing Your Part....22

Your Thyroid— The Basics

Your thyroid is a gland located at the base of your neck, just below your Adam's apple. It is shaped like a butterfly—each "wing," or **lobe**, of your thyroid lies on either side of your windpipe.



The purpose of your thyroid gland is to make, store, and release **thyroid hormone** into your blood. Thyroid hormone affects essentially every cell in your body, and helps control your body's functions. If you have too little thyroid hormone in your blood, your body slows down. This is a condition called **hypothyroidism**. If you have too much thyroid hormone in your blood, your body speeds up. This is a condition called **hyperthyroidism**.

For more information on the thyroid gland, please contact:

The Thyroid Foundation of America
Ruth Sleeper Hall, RSL 350
40 Parkman Street
Boston, MA 02114-2698
(send self-addressed, stamped envelope)

The Thyroid Society for Education & Research
7515 S. Main Street, Suite 545
Houston, TX 77030
1-800-THYROID

National Graves' Disease Foundation
2 Tsitsi Court
Brevard, NC 28712

Or visit our web site at:
www.glandcentral.com

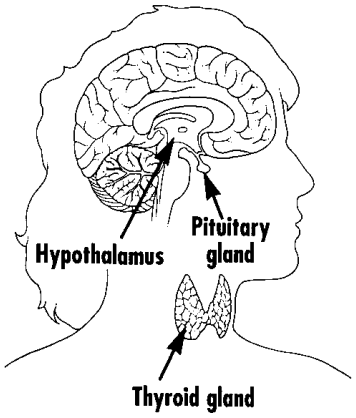
A Final Note

While finding out that you have thyroid cancer may seem overwhelming, there are very effective treatments available to you. Find out as much as you can about your condition so you can communicate with your doctor and address any concerns you may have in deciding which treatment may best fit your particular condition. With your doctor's expertise and your cooperation, you can overcome your thyroid condition and lead a normal, healthy life.

The amount of thyroid hormone made by your thyroid gland is adjusted by another gland called the **pituitary**, as well as a part of your brain called the **hypothalamus**.

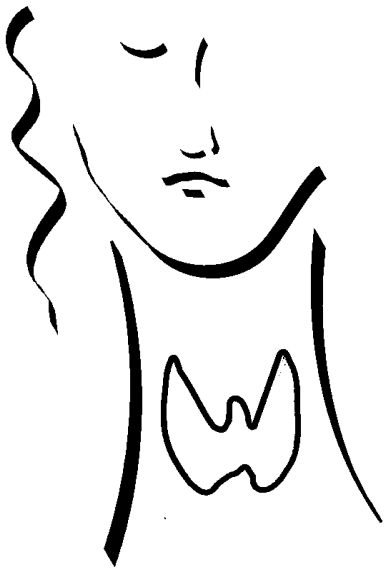
- The pituitary gland makes hormones that control many other glands in your body
- The hypothalamus helps the pituitary control many of your glands, as well as helping to control other bodily functions, such as thirst, hunger, sleep, and body temperature

The thyroid gland, pituitary gland, and hypothalamus all work together to control the amount of thyroid hormone in your body. With the pituitary controlling most of the action, these structures work in a way that is similar to the way a thermostat controls the temperature in a room.



Mild thyroid failure

A more mild form of hypothyroidism is **mild thyroid failure**. In this case, your thyroid hormone levels are normal, but you have a high TSH level. This may mean that TSH is making your thyroid gland work extra hard just to maintain normal levels of hormone in your blood. This condition often leads to hypothyroidism. People with mild thyroid failure generally don't show any obvious symptoms, which makes it very difficult for doctors to identify without performing a TSH test.



Of course, patients must continue to see their doctor for regular checkups to ensure that their TSH levels are in the proper range and that the dose of thyroid hormone medication is sufficient.

In addition, doctors will probably check their patients' blood **thyroglobulin levels** at the time of periodic checkups. Thyroglobulin is a protein that stores thyroid hormone in the body. It is made only by thyroid cells or thyroid cancer cells. In normal patients, thyroglobulin levels are usually very low. However, in patients with thyroid cancer, thyroglobulin levels may increase. A high thyroglobulin level may be a sign that the cancer has returned or spread.

Almost all patients who undergo thyroid surgery become hypothyroid and need to take thyroid hormone therapy to replace the thyroid hormone that their body can no longer produce on its own. In addition, many thyroid cancers may continue to grow under the influence of thyroid-stimulating hormone. Giving patients thyroid hormone tablets sends a signal to the pituitary gland to make less TSH—thereby limiting the growth of new cancer cells while keeping the proper level of TSH in the body.

Do patients ever relapse?

Relapses following the successful treatment of thyroid cancer are uncommon. This is because of the following reasons:

- Thyroid surgery often removes all of the cancer
- Radioactive iodine and X-ray therapy frequently kill any remaining cancer cells
- Thyroid hormone therapy helps prevent growth of new cancer cells

Hyperthyroidism

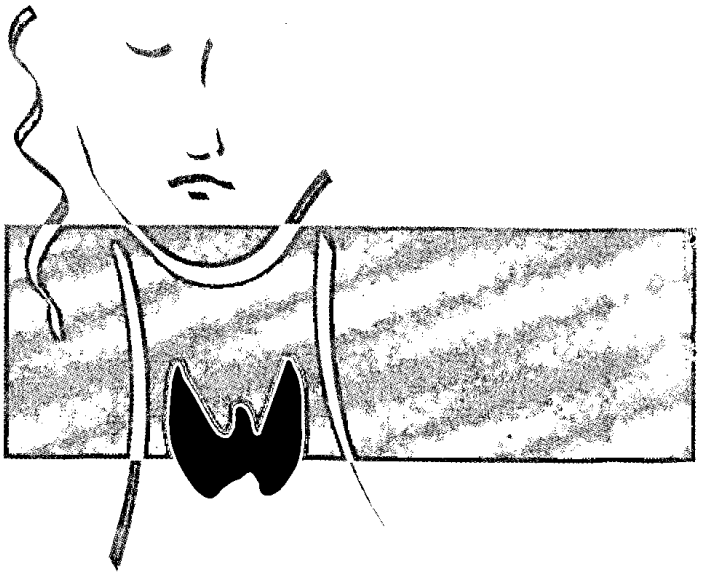
If your thyroid gland produces too much thyroid hormone, you may have a condition known as **hyperthyroidism**. If you are hyperthyroid, you also have a low level of TSH in your blood because your pituitary has decreased its TSH signals to your thyroid gland. Having too much thyroid hormone causes your body to speed up, resulting in nervousness, fast heartbeat, and unexplained weight loss, among other symptoms.

Mild (subclinical) hyperthyroidism

If your thyroid hormone levels are normal, but you have a low TSH level, you may have a condition known as **mild (subclinical) hyperthyroidism**. This often progresses to hyperthyroidism at a later point, and should be monitored carefully by your doctor.

Thyroid Cancer— A Closer Look

Thyroid cancer is a **malignant** (harmful) tumor or growth located within the thyroid gland. Thyroid cancer is uncommon, accounting for little more than 1% of all cancers in the US. If you or someone you know has been diagnosed with thyroid cancer, you will be glad to know that the outlook with treatment is excellent—most thyroid cancers can be **totally removed** with surgery and a **complete cure** is common.



Cancerous Thyroid Nodules

Thyroid cancer is very *rare*—only about 5% of thyroid nodules ever turn out to be cancerous. Furthermore, in most patients thyroid nodules grow **very slowly**, and the outlook with treatment is excellent—most thyroid cancers can be **totally removed** with surgery and a **complete cure** is common.

Treatment of thyroid cancer

Treatment of thyroid cancer usually begins with **surgery** to remove part or all of the thyroid gland. Thyroid surgery may be followed by treatment with radioactive iodine or X-ray therapy to attempt to destroy any "leftover" normal thyroid cells, as well as cancer cells.

Suspicious and Nondiagnostic Nodules

If your thyroid nodule is labeled "suspicious," this means that your doctor cannot say with certainty if the nodule is benign or cancerous. If this is the case, your doctor may rebiopsy the nodule, or recommend surgery to remove the nodule. Fortunately, most suspicious nodules turn out to be noncancerous.

If biopsy results are labeled "nondiagnostic," this means that there were not enough thyroid cells collected and a diagnosis cannot be made. Your doctor will probably repeat the thyroid biopsy.

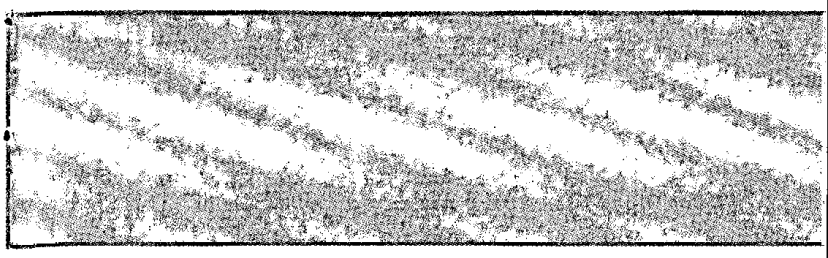
For more information, please refer to the booklet *Understanding Thyroid Nodules*.

Who Is at Risk?

Thyroid cancer is more likely to occur in people who have undergone radiation therapy of the head, neck, or chest during childhood. Radiation was commonly used before 1960 to shrink enlarged tonsils or adenoids, to treat various skin problems (such as acne), and to reduce an enlarged thymus gland (an organ inside the chest) in infants.

Thyroid cancer is also more likely to occur if you have a family member who has had thyroid cancer. However, thyroid cancer can occur in anyone.

Note: Radiation that is used in diagnostic X-rays (for example, X-rays used by dentists) is *not* connected with thyroid cancer.



Diagnosis of Thyroid Cancer

Thyroid cancer is often discovered by patients themselves. You may notice or feel a lump or **nodule** on the front of your neck, or your doctor may notice a nodule during a routine physical examination.

Symptoms of thyroid nodules may include difficulty in swallowing, hoarseness, or a cough. Sometimes thyroid nodules may be a sign that the thyroid gland is overactive—causing hyperthyroidism. As a result, patients may experience symptoms of an overactive thyroid, such as nervousness, rapid heartbeat, or unexplained weight loss.

Fortunately, most of the time, nodules are **benign**, which means they are not life-threatening. If fact, benign thyroid nodules are extremely common—it is estimated that they may be present in more than half of the US population. However, most nodules are so small that they are never discovered by the patient or doctor. Your doctor can usually make an accurate examination and diagnosis of a thyroid nodule with little risk or discomfort to you.

Treatment of Noncancerous Thyroid Nodules

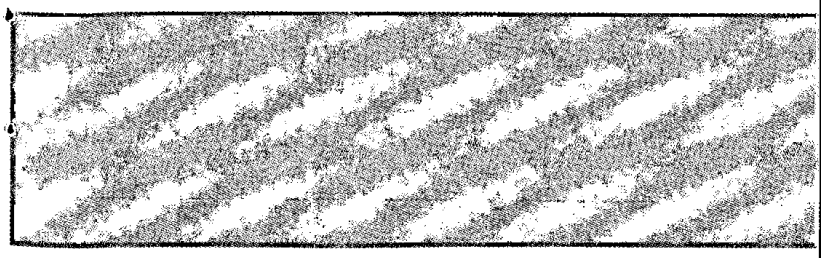
Your doctor may prescribe thyroid hormone tablets in an attempt to shrink the nodule. If you are treated with thyroid hormone tablets, you will need to see your doctor periodically for checkups so he or she can:

— Determine if the thyroid nodule has shrunk

— Check that your TSH level is in the proper range

— Adjust the dosage of thyroid hormone, if necessary

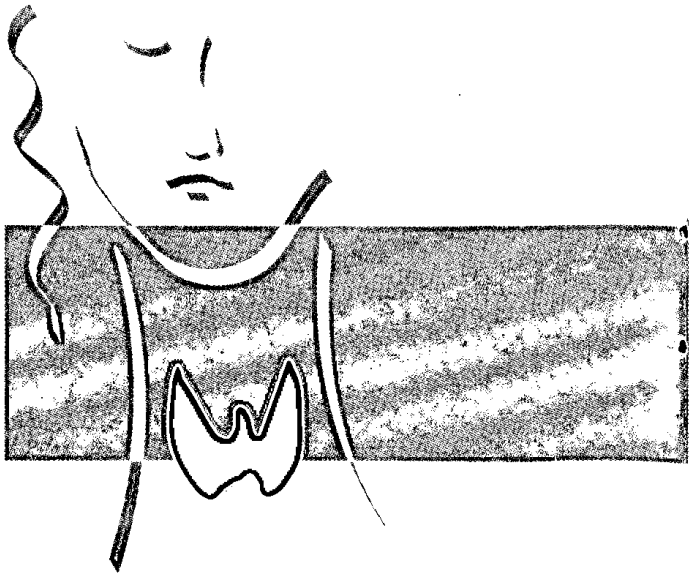
If the thyroid nodule does not decrease in size after several months of treatment, or if the nodule increases in size, your doctor may rebiopsy the nodule or recommend that the nodule be removed surgically.



The FNAB is usually performed in the doctor's office and takes about 10 to 20 minutes. The procedure is very safe, with most patients reporting only minimal discomfort. Biopsy results are usually available within a short time after the procedure (eg, within several days to a week).

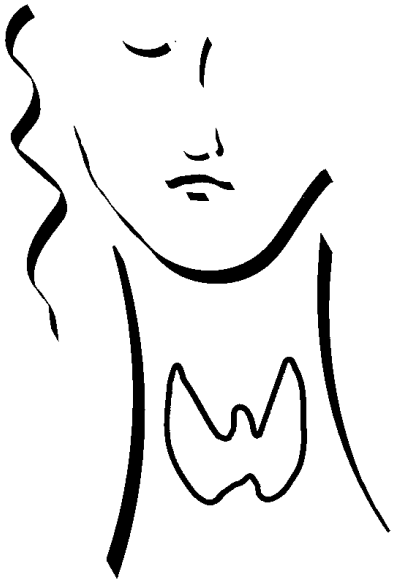
Based on the lab results of the FNAB, there are several ways that a thyroid biopsy can be interpreted:

- Benign or noncancerous
- Suspicious
- Nondiagnostic
- Cancerous or malignant



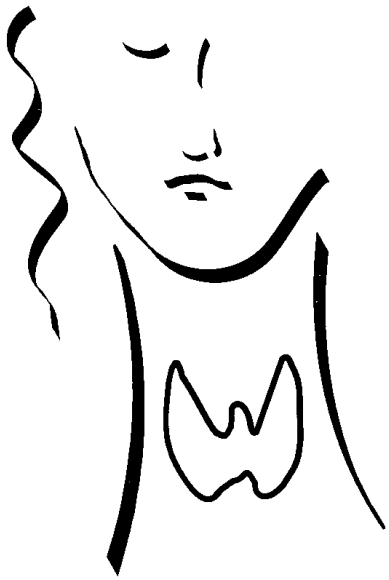
Once a thyroid lump or nodule is discovered, your doctor will take a careful medical history of your health. For example, your doctor may ask:

- Have you ever had radiation treatment of your head or neck?
- Does anyone in your family have thyroid disease?
- How long has the lump been there?
- Have you noticed a hoarseness or change in your voice?
- Do you have difficulty or discomfort when swallowing?



Your doctor will probably also ask you about how you have been feeling and whether you are experiencing any symptoms possibly caused by thyroid nodules. Your doctor will also conduct a physical examination to check for signs related to thyroid nodules. As part of this exam, your doctor will carefully examine your thyroid gland for a goiter or other nodules by pressing on it. (This procedure is called **palpation**.) After completing the physical examination of your thyroid, the doctor will likely conduct some additional tests, including:

- Blood tests (to measure the level of thyroid hormone)
- Thyroid imaging tests (to create a picture of your entire thyroid)
- Thyroid biopsy (to see if the lump contains cancer cells)

The importance of thyroid biopsy

A **fine needle aspiration biopsy (FNAB)**, or thyroid biopsy, can help determine whether a thyroid nodule is benign or cancerous in most patients. Many thyroid experts recommend an FNAB as the first test to do for a nodule. With this procedure, the skin may be numbed and a small needle is inserted into the thyroid nodule in order to remove samples of tissue or fluid, which are then analyzed in a lab.